

Cypress Testing: From Start to End

Automating Web Application Testing for Efficiency and Reliability

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What is Cypress?

Cypress is a modern JavaScript-based end-to-end testing framework. It automates web application testing, ensuring features like login, registration, and shopping carts work correctly before deployment.

Automated Testing

Runs multiple test cases quickly.

JavaScript-Based

Modern and easy to integrate.

End-to-End

Tests full user flows.

Installation Steps

01

Install Node.js

Required for Cypress to run.

03

Install Cypress

Run: `npm install cypress --save-dev`

02

Initialize Project

Create folder and run: `npm init -y`

04

Open Cypress

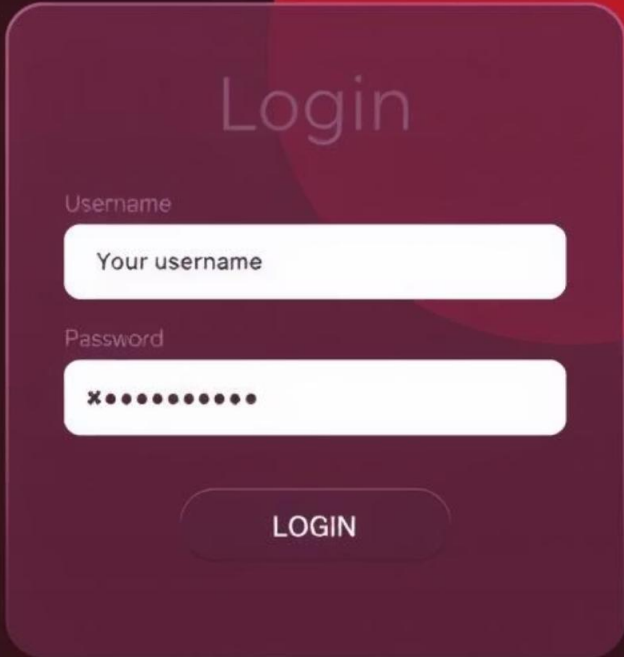
Execute: `npx cypress open`

Demo Example: Login Form Testing

We will test a sample login form from practicetestautomation.com.

Test Cases

- Valid Login: Correct username & password.
- Invalid Login: Wrong password.
- Invalid Login: Empty fields.



The image shows a login form titled "Login" on a dark purple background. The form is a lighter purple rounded rectangle. It contains two input fields: "Username" with the placeholder text "Your username" and "Password" with masked characters "x••••••••". Below the fields is a "LOGIN" button. A large red circle is visible in the background behind the form.

Microsoft Windows [Version 10.0.26100.4946]
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C:\Users\anant>cd cypress-demo

C:\Users\anant\cypress-demo>npx cypress open|

Cypress Test Code Example

```
describe('Login Form Tests', () => {  
  it('Valid login with correct credentials', () => {  
    cy.visit('https://practicetestautomation.com/practice-test-login/')  
    cy.get('#username').type('student')  
    cy.get('#password').type('Password123')  
    cy.get('#submit').click()  
    cy.contains('Logged In Successfully').should('be.visible')  
  })  
})
```

This code snippet demonstrates a valid login test case, visiting the URL, typing credentials, clicking submit, and asserting success.

How Cypress Checks Automatically

Assertions

Cypress uses assertions like

```
.should('be.visible')
```

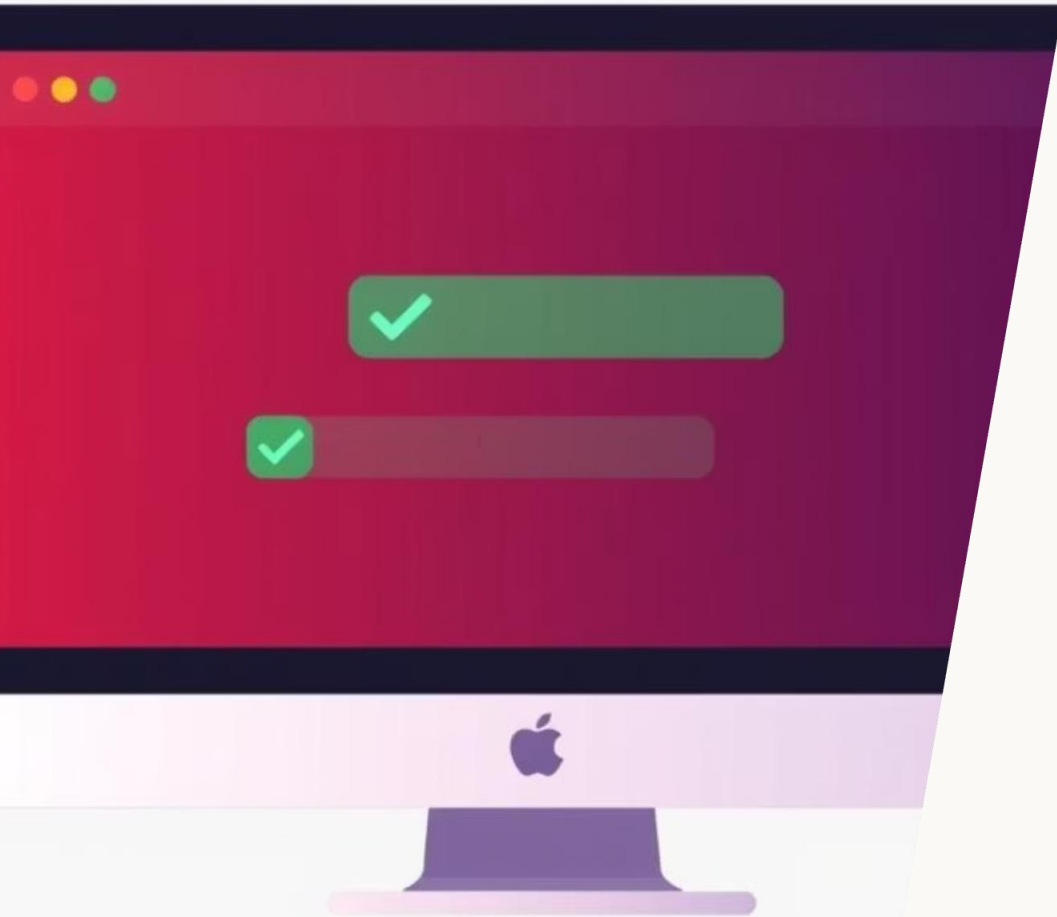
to check conditions.

Pass/Fail Indicators

Tests are marked **PASS** (green) if conditions pass, or **FAIL** (red) if they fail.

Debugging Aids

Failed tests provide screenshots and video recordings for easy debugging.



Real-World Usage

Cypress tests are crucial in real software companies for continuous integration and delivery.



CI/CD Integration

Integrated with pipelines like GitHub, Jenkins, GitLab.



Automated Runs

Tests run automatically when new code is pushed.



Ensures Stability

Prevents breakage of critical features after updates.



Faster Releases

Leads to quicker releases with fewer bugs and happier customers.

Advantages of Cypress Testing Tool



Seamless In-Browser Execution

Tests run directly alongside your app, eliminating remote driver overhead for speed and reliability.



Effortless Automation

Automatic waiting and real-time reloads provide instant feedback, accelerating your development workflow.



Superior Debugging

Time-travel, clear errors, and browser dev tool integration make troubleshooting intuitive and efficient.



Robust Control

Stub network requests and isolate tests for thorough edge case testing without backend dependencies.



Visual Evidence

Built-in screenshots and video recording simplify post-failure analysis in CI pipelines.



Developer-Friendly

Leverages the modern JavaScript ecosystem with extensive documentation and a supportive community.

Disadvantages of Cypress Testing Tool



JavaScript Only

Limited to JavaScript, unlike Selenium's multi-language flexibility.



Single Window Focus

Cannot handle multi-tab or multi-window scenarios, restricting complex workflows.



Limited Browser Coverage

No support for Safari or Internet Explorer, impacting broader cross-browser testing.



Web-Only

Exclusively for web apps; not suitable for native mobile app testing.



Costly Parallelization

Resource-intensive parallel testing, often requiring multiple machines for CI.



Advanced Learning Curve

Mastering advanced features like custom commands requires deeper JavaScript expertise.

Conclusion & Real-World Impact

Cypress automates repetitive testing, saving time and reducing errors. By verifying critical features, it ensures software quality.



Time Savings

Automates repetitive tasks.



Error Reduction

Minimizes human error in testing.



Widely Used

Essential for modern web app testing.

Just like testing our mini-Student Portal, real-world applications like Flipkart use hundreds of Cypress tests to ensure seamless user experience.